

Piecewise linear graphs



1

a $a = -1$

b -2

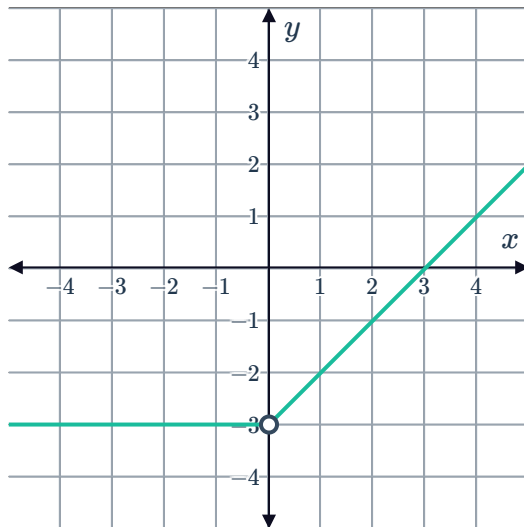
2

a
$$y = \begin{cases} \frac{3}{2}x + 3 & \text{if } x \leq 0 \\ 3 - x & \text{if } 0 < x \leq 3 \\ 2x - 6 & \text{if } 3 < x < 5 \end{cases}$$

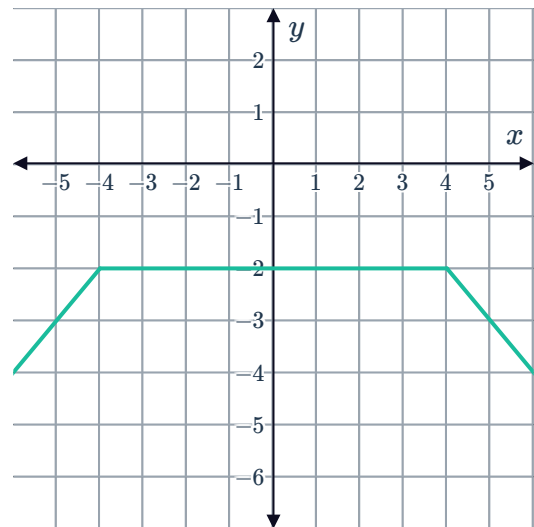
b
$$y = \begin{cases} x + 1 & \text{if } -5 \leq x \leq 1 \\ 1 - x & \text{if } 1 < x \leq 4 \end{cases}$$

3

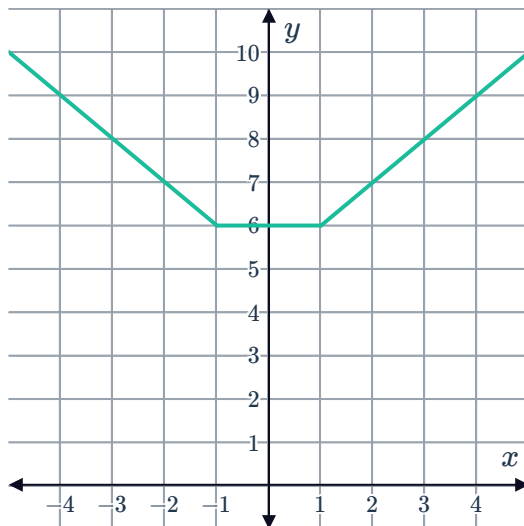
a



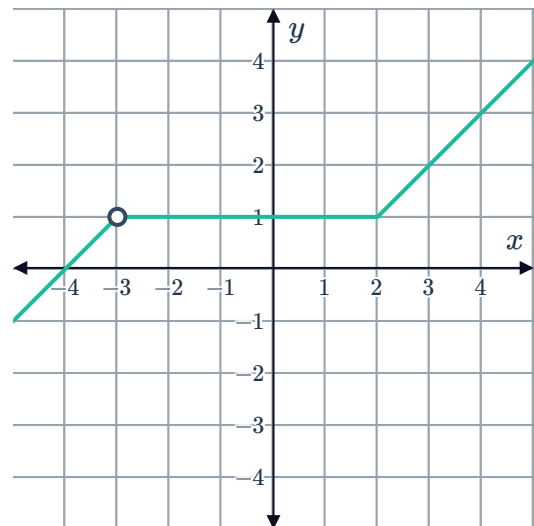
b



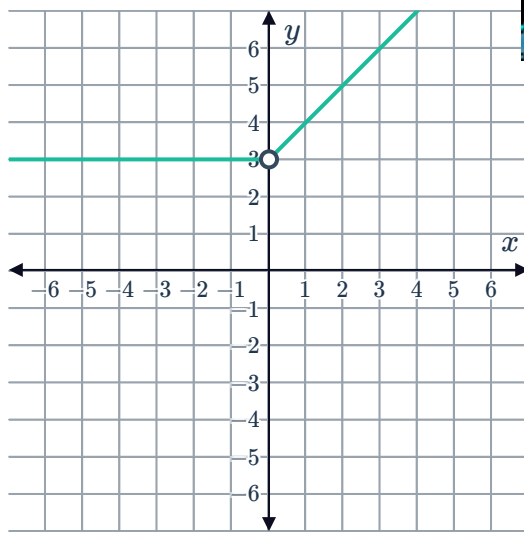
c



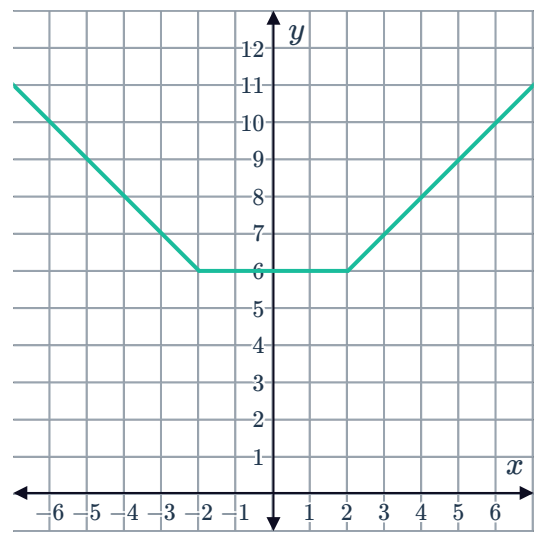
d



e



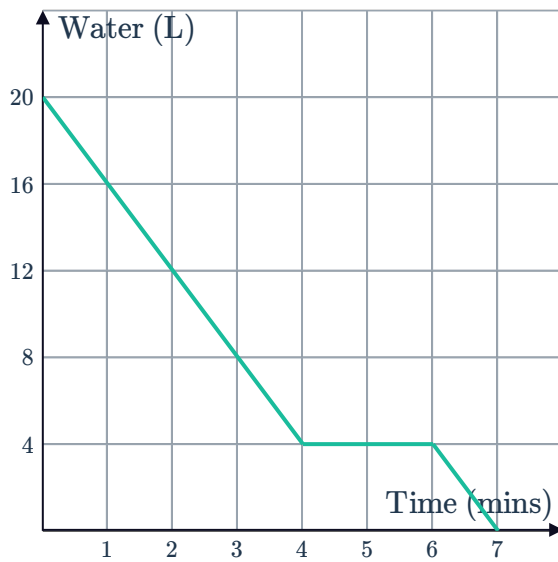
f



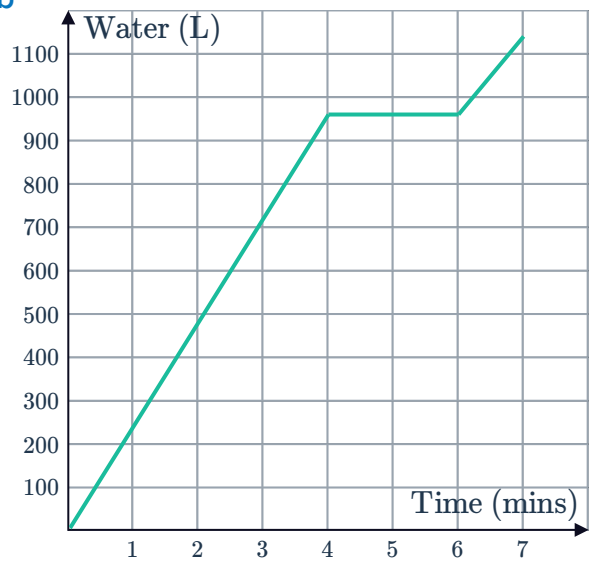
Piecewise Models

4

a



b



5

a 16 L

c 16 L

b The driver filled the tank.

d 12 pm

6

a 2 km

b 1 pm

c 4 km

7

a 2.5 km

b 3:00 pm

c 3.5 km

8

a 200 L

b $V = \frac{40}{9}t$

c $t = 95$

9

a
$$D = \begin{cases} 0.25t & \text{when } 0 \leq t < 40 \\ 2 + 0.2t & \text{when } 40 \leq t \leq 200 \end{cases}$$

b

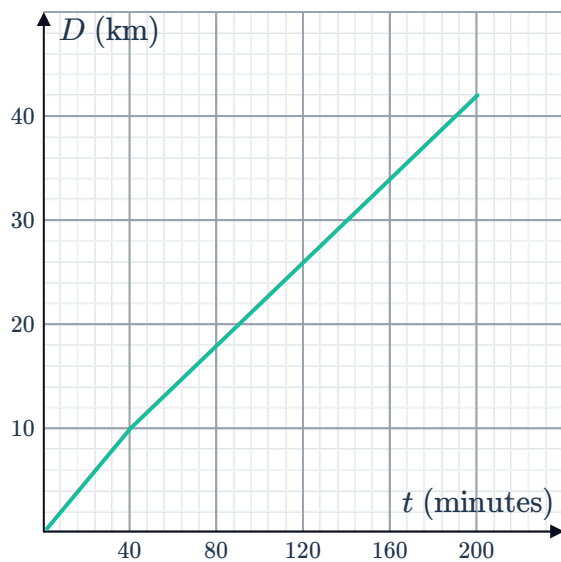
i 5 km

ii 10 km

iii 14 km

iv 42 km

c



10

a

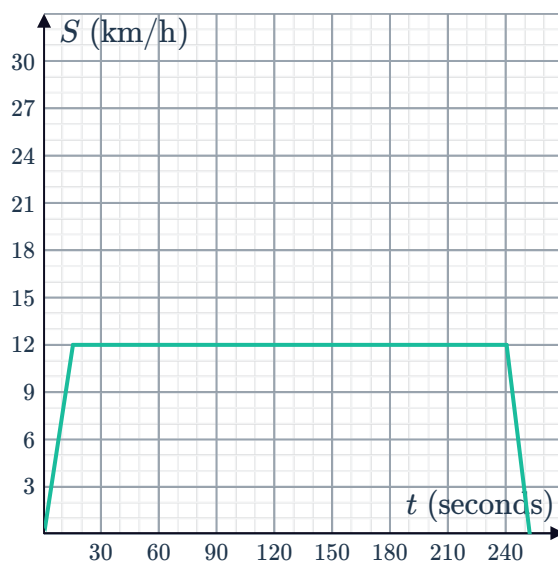
i 8 km/h

ii 12 km/h

iii 12 km/h

iv 2 km/h

b



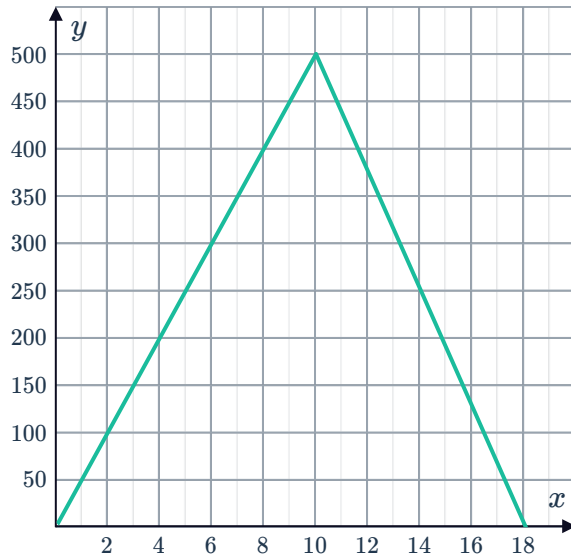
11

a 12

b $E = 14$ c $24 < t \leq 36$

12

a



b

748 m

Step graphs

13

a $y = 1$

b $y = -3$

14

a $y = -5$

b $y = 8$

15

a $y = 3$

b $y = -4$

Step graph models

16

a

i \$1

ii \$5.50

b 175 g

17

a \$10

b 4.75 hours

c \$6

18

a \$9

b \$12

c 5 hours

19



20

- a 1 hour b \$5 c \$6

21

- a \$2400 b \$2000 c 1 hour

22

- a \$3 b \$3 c 15

23

- a \$1.70 b \$1.30 c 4 minutes d \$0.20
e \$0.70

24

- a
i \$10 ii \$12 iii \$6 iv \$14
b Less than 2 kg
c
i Less than 1 kg ii Less than 1 kg
iii Less than 0.5 kg iv Less than 4 kg

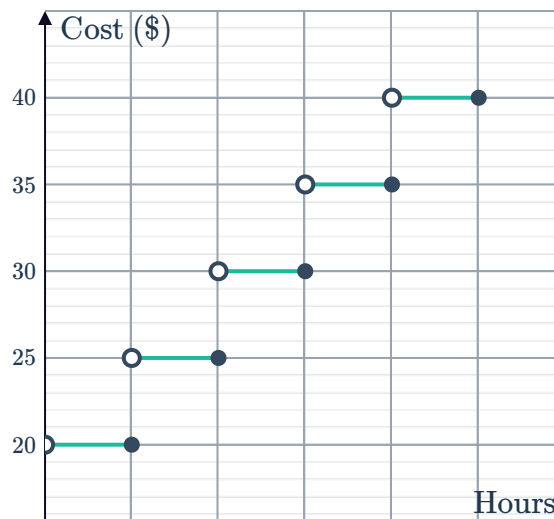
25

- a \$2.50 b \$8 c \$1.50 d 16 km
e \$55

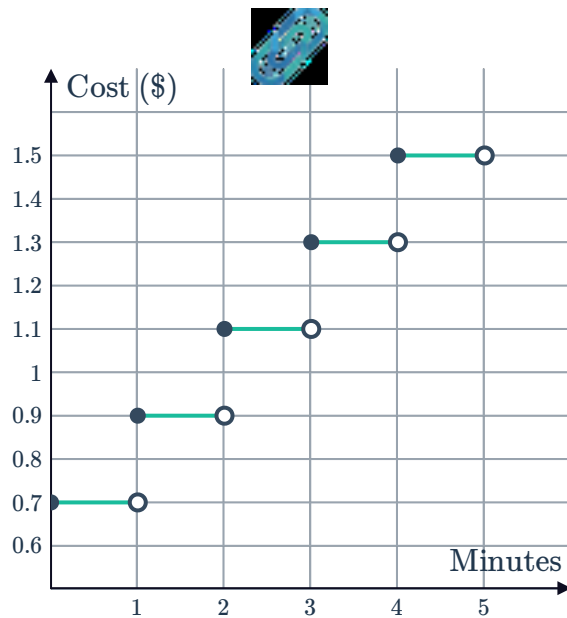
26

- a \$2.10 b \$6 c \$51.30

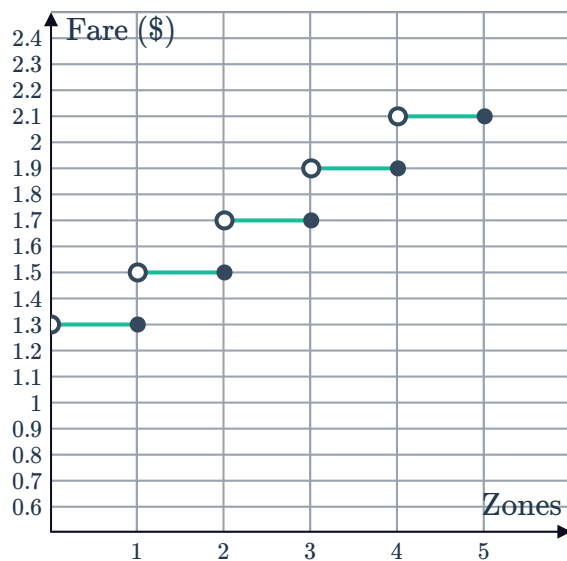
27



28



29

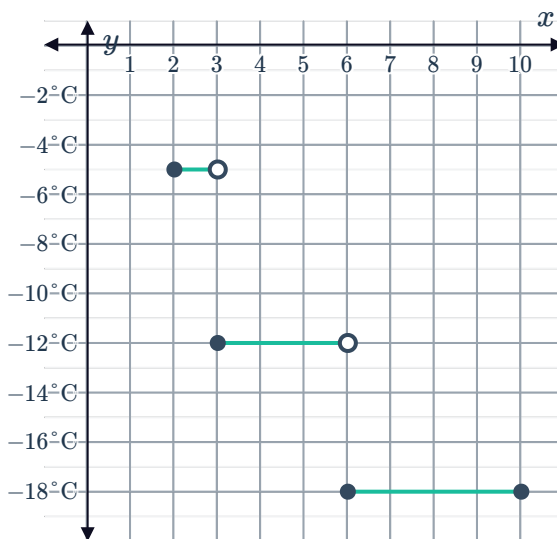


30

a

$$y = \begin{cases} -5^{\circ}\text{C}, & 2 \leq x < 3 \\ -12^{\circ}\text{C}, & 3 \leq x < 6 \\ -18^{\circ}\text{C}, & 6 \leq x \leq 10 \end{cases}$$

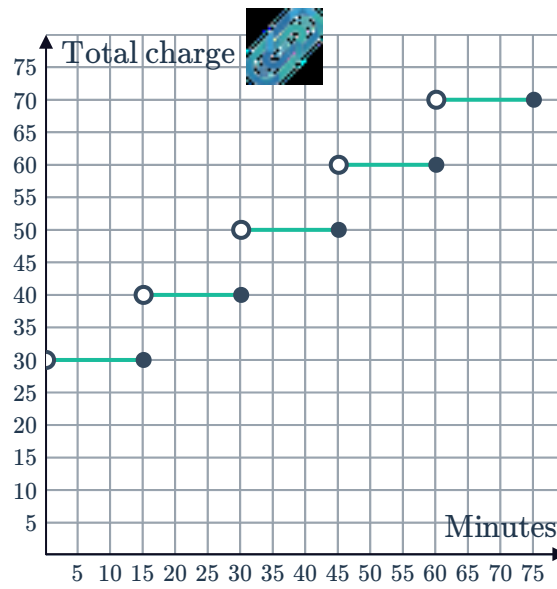
b



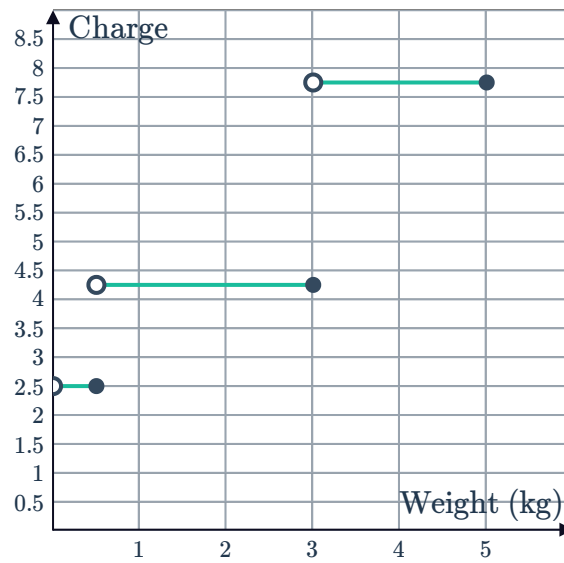
c -12°C

d 3 pm

31



32



33

a \$1200

b \$48 999

c \$1200